



OASIS Security Report

Sensitive Data Exposure Security Analysis

Date: 2026-06-11 10:02:19 — **Model:** mistral:latest

Vulnerability: Sensitive Data Exposure

Project: clean_contracts

Analysis root: ../clean

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Summary

Analyzed **2** file(s) for Sensitive Data Exposure.

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Statistics

Files analyzed: 2 **Total findings:** 2

Critical		0
High		0

Medium**2****Low****0**[Return to table of contents](#)

Files

File	Similarity
<code>/work/project/dataset/clean/simple_dao_fixed.sol</code>	0.600
<code>/work/project/dataset/clean/modifier_reentrancy_fixed.sol</code>	0.593

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Detailed analysis

File 1: /work/project/dataset/clean/simple_dao_fixed.sol

Similarity score: 0.600

Source lines: 1-34

► Finding 1.1: Vulnerability found (Medium)

Vulnerable code

```
credit[msg.sender]>= amount
```

Explanation

The contract stores credit values in plaintext, exposing the balance of each address.

Impact

Exposure of user balances

Entry point

withdraw function

Execution path

```
'''  
start --> withdraw(amount) --> if (credit[msg.sender]>= amount) -->  
credit[msg.sender]>= amount ----> exposure  
'''
```

Exploitation steps

- Call the withdraw function
- Inspect exposed credit balance

Remediation

Store sensitive data encrypted, or use a secure storage solution to protect the user balances.

File 2: /work/project/dataset/clean/modifier_reentrancy_fixed.sol

Similarity score: 0.593

Source lines: 1-42

► Finding 2.1: Vulnerability found (Medium)

Vulnerable code

```
tokenBalance[msg.sender] == 0
```

Explanation

Hardcoded public contract name ('Nu Token') is exposed in the Bank contract function.

Impact

Exposure of contract name, which could aid attackers in identifying and targeting vulnerable contracts.

Entry point

Bank.supportsToken() external function

Execution path

```

    ...
    Bank.supportsToken()
    |
    +---> ModifierEntrancy.
    ...

```

Exploitation steps

- Identify contract name from exposed data.
- Target vulnerable contracts with the identified name.

Conditions

Attacker has knowledge of contract names in use.

Remediation

Store hardcoded secrets securely, e.g., as off-chain constants or using a secure configuration management system.

Secure example

```
const PRIVATE_KEY = '0xYOUR_PRIVATE_KEY'; // Store your private key securely.
```

```
function supportsToken() internal returns(bytes32) {  
    require(keccak256(abi.encodePacked(PRIVATE_KEY)) == bank.supportsToken());  
}
```

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Assistant (triage / codebase)

In the OASIS dashboard (JSON report preview), use the chat below to triage findings, exploitation paths, and remediation. Select file, chunk, and finding to narrow context when needed.

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Errors & Notes

No file-level errors were recorded.

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